

Elgamal Digital Signature

Elgamal Digital Signature facts:

- A modified version is used in the NIST Digital Signature Standard (DSS)
- Uses a secure hash algorithm (SHA)

The sender has the public key (p, g, y) , where:

- p is a large prime
- g is a primitive root
- $y = g^x \bmod p$

The sender has the private key (k, x) to sign the message, where:

- k is a random integer such that $1 < k < p-1$ and $\text{GCD}(k, p-1) = 1$
- x is difficult to compute for large values of p

To sign a message m , where $0 \leq m \leq p-1$:

- Get a random integer k , where $1 < k < p-1$ and $\text{GCD}(k, p-1) = 1$
- Get the hash of m , $h(m)$
- Compute: $s_1 = g^k \bmod p$
- Compute: $k^{-1} \bmod (p-1)$
- Compute: $s_2 = k^{-1} (h(m) - xs_1) \bmod (p-1)$
- Send the message m and the signature (s_1, s_2) as: $m, (s_1, s_2)$

The receiver verifies by computing v_1, v_2 , and checking if $v_1 = v_2$ where:

- $v_1 = g^{h(m)} \bmod p$
- $v_2 = y^{s_1} s_1^{s_2} \bmod p$

This works (unbelievably!) because:

$$\begin{aligned}
 v_1 &= v_2 \\
 g^{h(m)} \bmod p &= y^{s_1} s_1^{s_2} \bmod p \\
 g^{h(m)} \bmod p &= (g^x)^{s_1} (g^k)^{k^{-1}(h(m)-xs_1)} \bmod p \\
 g^{h(m)} \bmod p &= (g^{xs_1}) (g^{kk^{-1}(h(m)-xs_1)}) \bmod p \\
 g^{h(m)-xs_1} \bmod p &= g^{kk^{-1}(h(m)-xs_1)} \bmod p
 \end{aligned}$$

$$h(m) - xs_1 \bmod (p-1) = kk^{-1}(h(m) - xs_1) \bmod (p-1)$$

$$h(m) - xs_1 \bmod (p-1) = h(m) - xs_1 \bmod (p-1)$$

We have used the fact that $g^i \equiv g^j \bmod p$ iff $i \equiv j \bmod (p-1)$

Example:

Sender has public key $(p=19, g=10, y=4)$, and private key $(k, x=16)$.

- Chose random integer $k=5$, where $1 < k < 18$, $\text{GCD}(5,18)=1$
- Get hash of m , $h(m)=14$
- Compute: $s_1 = 10^5 \bmod 19 = 3$
- Compute: $5^{-1} \bmod (19-1) = 5^{-1} \bmod (18) = 11$
- Compute: $s_2 = 11(14 - 16(3)) \bmod (19-1) = -374 \bmod 18 = 4$
- Send the message and the signature: $m, (3, 4)$

The receiver verifies by computing v_1 , v_2 , and checking if $v_1 = v_2$ where:

- $v_1 = 10^{14} \bmod 19 = 16$
- $v_2 = (4)^3 (3)^4 \bmod 19 = 5184 \bmod 19 = 16$

Exercise:

Akuma has Elgamal public key $(p = 13, g = 6, y = 3)$.

1. Verify the message sent from *Akuma* with hash $h(m) = 10$ and signature $(s_1 = 2, s_2 = 6)$.
2. Forge *Akuma's* signature on the message m with hash $h(m) = 8$ using $k = 5$.
3. Verify the forged signature from question 2.

Answers:

1. $v_1 = 6^{10} \bmod 13 = 4$
 $v_2 = 3^2 2^6 \bmod 13 = 4$
2. $x = 8$
 $m, (s_1 = 2, s_2 = 8)$
3. $v_1 = 6^8 \bmod 13 = 3$
 $v_2 = 3^2 2^8 \bmod 13 = 3$